

Diagnosis of attention deficit hyperactivity disorder using non-linear analysis of the EEG signal

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Yasaman Kiani Boroujeni¹, Ali Asghar Rastegari¹, Hamed Khodadadi² ✉

¹Department of Molecular and Cell Biochemistry, Falavarjan Branch, Islamic Azad University, Isfahan, Iran

²Department of Electrical Engineering, Khomeinishahr Branch, Islamic Azad University, Isfahan, Iran

✉ E-mail: khodadadi@iaukhsh.ac.ir

Abstract: Attention deficit hyperactivity disorder (ADHD) is a common behavioural disorder that may be found in 5%–8% of the children. Early diagnosis of ADHD is crucial for treating the disease and reducing its harmful effects on education, employment, relationships, and life quality. On the other hand, non-linear analysis methods are widely applied in processing the electroencephalogram (EEG) signals. It has been proved that the brain neuronal activity and its related EEG signals have chaotic behaviour. Hence, chaotic indices can be employed to classify the EEG signals. In this study, a new approach is proposed based on the combination of some non-linear features to distinguish ADHD from normal children. Lyapunov exponent, fractal dimension, correlation dimension and sample, fuzzy and approximate entropies are the non-linear extracted features. For computing, the chaotic time series of obtained EEG in the brain frontal lobe (FP1, FP2, F3, F4, and Fz) need to be analysed. Experiments on a set of EEG signal obtained from 50 ADHD and 26 normal cases yielded a sensitivity, specificity, and accuracy of 98, 92.31, and 96.05%, respectively. The obtained accuracy provides a significant improvement in comparison to the other similar studies in identifying and classifying children with ADHD.

1 Introduction

Attention deficit hyperactivity disorder (ADHD) is known as a mental and behavioural disorder, develops from childhood into adulthood [1]. The symptoms of ADHD are by excessive activity, problems in paying attention, restlessness, and sudden behaviours [2]. Recent studies performed on different populations have indicated that ADHD is observed in about 5% of children, with more frequency in boys [2, 3]. In addition, this disease has some effects such as poor inhibitory control on adults with ADHD [4] and affects the performance of their working memory [5].

Currently, there is a consensus that some sophisticated connection systems, such as neuro-anatomical and neurochemical systems, are the main causes of hyperactivity disorder [6, 7]. Early diagnosis of ADHD has an important role in preventing subsequent problems in children's social interactions [8] and can help the children, parents, and the community health [9]. Usually, criteria of Diagnostic and Statistical Manual of Mental Disorders (DSM) or International Classification of Diseases (ICD) are used to diagnose ADHD in clinics [2]. The high dependency of children's response to the parents' interpretation of the psychologist's questions and the degree of honesty in the children's answers are some of the challenges in the diagnosis of ADHD. For reducing the mentioned problems, the computers can be utilised by physicians for diagnosing specified diseases. This approach is named computer-aided design (CAD) in which, medical images or physiological signals are analysed automatically for diagnosing the diseases [10].

Several approaches are proposed based on electroencephalogram (EEG) signal analysis in the literature for diagnosing the ADHD [11–14]. The first abnormality study on EEG signals for ADHD diagnosis was performed in [15]. It was proved that some of the indices extracted from EEG signals could show the ADHD abnormality and learning disabilities [14]. The study performed in [16] indicated that the EEG frequency content for normal and ADHD cases, appears in the relative power of standard EEG bands are completely different. The brain maps of ADHD children have lower bands in alpha and beta, and higher bands in theta and delta compared to the normal ones. It is used to identify ADHD cases [9, 13]. EEG signal is one of the significant

tools utilised to diagnose the neurological diseases and clinical evaluation of brain activity [17–19].

The brain has a highly non-linear dynamic system and its activity shows a chaotic behaviour [20]. Modelling the brain dynamics and detecting the variations in the EEG signals could be performed by the non-linear dynamics and chaos theory [21]. In other words, chaos theory is used to EEG signal analysis for diagnosing medical disorders and diseases. The application of chaotic dynamics for studying biological systems and signals has increased recently [22]. The surveyed studies indicate that chaos theory plays an important role in neural system analysis, and extracting the brain signal properties [23]. In addition, [24] shows that the non-linear analysis of EEG signals can investigate the neural dysfunctions in ADHD children. The non-linear features applied in some similar studies for EEG signal processing proved the chaotic pattern in the EEG signals [25, 26]. The most important non-linear features reported in literature are as follows: fractal dimension (FD) (which is calculated based on Higuchi, Katz, and Petrosian methods) [14, 25], largest Lyapunov exponent (LE) [27], approximate entropy (ApEn) [14], fuzzy entropy (FuzzyEn) [28], multiscale entropy [3, 29], Lempel-Ziv complexity, correlation dimension (CD), and Hurst exponent [30]. In the non-linear analysis performed on EEG signal, the focus is on diagnosis epilepsy, Alzheimer and ADHD, Autism spectrum disorder, estimating the depth of anesthesia and differentiating between various mental states in normal EEG signals [31].

On the other hand, although the chaotic behaviour of brain activity and therefore its generated EEG signals are proved, the chaotic indices have not widely applied in the diagnosis of ADHD. Therefore, this paper tries to extract the chaotic indices for the EEG signal, combine them to create a diagnostic criterion and use this factor for diagnosing and classifying the ADHD/normal participants. Various methods have been proposed to quantify the chaos in which, LE, FD, CD, sample entropy (SampEn), FuzzyEn and ApEn are the most important features. It should be noted that each of the calculated non-linear indices shows an aspect of the chaotic behaviour of the EEG signals. Sensitivity to the initial condition, pattern complexity, entropy production rate and irregularity of a signal can be measured by the introduced features. Therefore, the combination of these features can illustrate the



Fig. 1 EEG data acquisition in three modes of (a) EO, (b) EC, (c) CT



Fig. 2 EEG signals in eyes closed condition for two cases (a) An ADHD patient, (b) A normal case

various aspects of the chaotic nature of the EEG signals in ADHD cases.

In the present study, all of the above-mentioned indices are calculated on a set of EEG signal obtained from 50 ADHD and 26 normal children. The experiments are performed on EEG signals obtained from five channels of the frontal lobe (consisted of FP1, FP2, F3, F4, and Fz) during a Continuous Performance Test (CPT). To evaluate the chaotic nature of ADHD, the obtained features are combined together. The ability of the combination of the chaotic indices is also assessed in making a distinction between ADHD cases and healthy children. Besides, the EEG signal data set is classified through the SVM approach in terms of the computed features. To assess the classification precision, three evaluation terms i.e. sensitivity, specificity, and accuracy are calculated. A comparison between the obtained results in this study and other similar works, which have used the EEG analysis for ADHD

detection, evinced that the selected combination of chaotic indices overcomes the other methods in the ADHD diagnosis accuracy.

The paper is structured as follows. The obtained data sets consist of 76 EEG signal are introduced in Section 2. Besides, the chaotic indices description and their calculation method are presented in the same section. Section 3 provides the results of experiment EEG signal processing, the selecting feature method as well as the classification and comparison. Section 4 includes discussion. Finally, the conclusion is presented in section 5.

2 Materials and methods

2.1 EEG data set

The data set employed in this paper was obtained from 50 children with ADHD (37 boys and 13 girls) and 26 normal children (16 girls and 10 boys) aged 4 to 15 years old, who had come to doctor Mohammad Behdad (neurologist) clinic for EEG signal recording in Isfahan, Iran. A neurologist analyzed the obtained EEG time series signals and diagnosed the ADHD individuals based on DSM-IV criteria. EEG signals were recorded via the Mitsar using WinEEG software with a sampling rate frequency of 256 Hz and 16-bit resolution [12]. After the individuals sat in a silent room comfortably, the Data recording was performed for at least 9 to 10 min in three equal modes consist of Eyes Open (EO), Eyes Closed (EC) and Cognitive Tasks (CT). The data acquisition setup for these three modes is shown in Fig. 1.

The placement of EEG electrode was compatible with the international 10/20 system applying an electrocap. In this system, the brain activities are recorded in 19 regions from Fp1, Fp2, F3, F4, F7, F8, Fz, C3, C4, Cz, T3, T4, T5, T6, P3, P4, Pz, O1 and O2, referenced to link ears [24]. A1 and A2 electrodes are placed on earlobes. For minimising the noise and artifacts, the participants were asked to minimise their movement particularly in their head area [15]. In this way, ten seconds of time series signals were selected for analyzing and diagnosing regarding the Neuroguide software criteria and physician's opinion. This selection makes the visual inspection of the pre-processing stage, is performed to reduce the computational demand and noise effects on signals. In addition, the collected EEG signals were passed with several filters consist of a lowpass-filter with 50 Hz and a highpass-filter with 1 Hz cut-off frequency to eliminate the artifacts. Furthermore, the EEG signals are broadband-filtered between 2 and 45 Hz through a fifth-order Butterworth filter.

Therefore, ten seconds artifact-free time series signals were obtained for any individuals after both sections of the pre-processing stage [12]. In Fig. 2, the recorded EEG signals are presented in the two ADHD and normal cases in eyes closed condition. The selected times depicted in the images illustrate signals without noise and artifacts. Fig. 3 shows EEG signals for F3 channel in the two cases of ADHD and normal in eyes closed condition.

2.2 Feature extraction

Chaos theory is one of the powerful tools for discovering the mathematical biomarkers for various pathologies fields [32]. In this section, the selected chaotic indices are described and the methods of their calculations are presented.

2.2.1 Lyapunov exponents: The rate of exponential divergence for two nearby trajectories can be quantified by the LE. It also can evaluate the chaos degree of a system [33]. If at least one positive LE has appeared in the LE spectrum of a time series, the chaos has emerged. For a one-dimensional dynamical system ($X_{n+1} = F(X_n)$), LE can be estimated as [32]

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln \left| \frac{dF(x)}{dx} \right|_{x=x_i} \quad (1)$$

For a multidimensional system, the number of LEs is as the same as the system dimension. Several methods such as Wolf algorithm have been employed for LEs calculation in time series in the

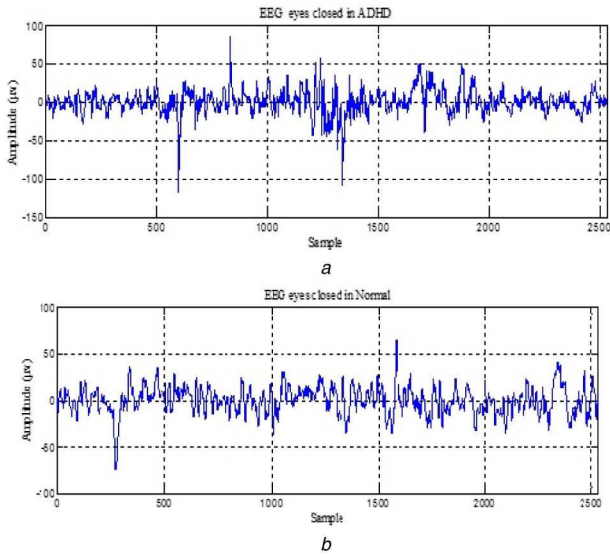


Fig. 3 Selected section of EEG signals for F3 brain area in eyes closed condition for two cases
(a) An ADHD patient, (b) A normal case

literature. However, these approaches calculate the largest LEs [34]. The method of this study for LEs calculation is similar to [32]. In this method, through the Time Delay Estimation (TDE) approach, the system phase space is reconstructed and named as the Reconstructed Phase Space (RPS).

2.2.2 Phase-space reconstruction using TDE: RPS is one of the effective methods for analyzing the chaotic time series. In this method, via the available time series, the system state space is reconstructed. If all of the system dynamics is not accessible, the precise phase space is not available by using a time series [32]. Hence, an RPS is built which is a pseudo phase space of the system.

Let x_i represents a chaotic time series ($1 < i < N$). By considering the embedding dimension as m as well as the time delay as τ , the system is embedded in its delayed RPS. The reconstructed delay phase space points are as follows:

$$X_i = (x_i, x_{i+\tau}, \dots, x_{i+(m-1)\tau})^T \quad (2)$$

The principle of TDE approach is based on using some scalar data to reconstruct an N -dimensional phase space. For employing this approach, two important parameters of m and τ should be computed. The RPS matrix is obtained as:

$$X = \begin{bmatrix} x_1 & x_{1+\tau} & \cdot & \cdot & \cdot & x_{1+(m-1)\tau} \\ x_2 & x_{2+\tau} & \cdot & \cdot & \cdot & x_{2+(m-1)\tau} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ x_m & x_{m+\tau} & \cdot & \cdot & \cdot & x_{m+(m-1)\tau} \end{bmatrix} \quad (3)$$

In this paper, the Jacobian method is utilised for the LEs calculation. Therefore, the linearisation is performed for a small variation near the system trajectory. Let the state space model of the system described as:

$$X_{n+1} = F(X_n), \quad n = 0, 1, \dots \quad (4)$$

Linearisation of the system yields:

$$X_{n+1} \cong J_n \times X_n, \quad n = 0, 1, \dots$$

$$J_n = \left. \frac{\partial F}{\partial X} \right|_{X_n} \in R^{n \times n}. \quad (5)$$

Considering the state variable iteration of the system in a $\{X_1, X_2, \dots, X_n\}$ sequence form yields to the Jacobian matrixes sequence of $\{J_0, J_1, \dots, J_{n-1}\}$. Y^n is obtained by multiplication of the Jacobian matrixes as:

$$Y^n = J_{n-1} \times J_{n-2} \times \dots \times J_1 \times J_0 \quad (6)$$

by defining the symmetric positive definite matrix as:

$$\Lambda = \lim_{n \rightarrow \infty} ((Y^n)^T Y^n)^{1/2n}. \quad (7)$$

the system LEs can be defined as the logarithms of eigenvalues of Λ [34].

2.2.3 Fractal dimension: FD as one of the chaos measurement tools is widely utilised in the determination of chaotic behaviour of signal and images. Box-counting method is one of the most used approaches for the FD calculation. In this method, a network of squares is created on the contour, and the number of squares in this network that includes a part of the curve is calculated. The size of the squares changes, then, the occupied squares are counted again. A point is obtained by calculating the logarithm of the number of counted squares and the logarithm for the reduction coefficient. New points are obtained by reducing the size of the network, which is connected in a curve. Calculating the slope of this curve indicates the FD [35].

Equation (8) describes how the self-similarity dimension (D), the number of self-similar pieces (a) and the reduction factor ($1/S$) are related together.

$$a \propto \frac{1}{S^D} \quad (8)$$

where D is defined as:

$$D = \frac{\log(a)}{\log(1/S)} \quad (9)$$

D can be calculated by estimating the slope of the approximated line for the plot of $\log(a)$ vs. $\log(1/S)$ [35]. There are several approaches for estimating FD in which, Box Counting Methods (BCM) is used in this study.

The more uneven pattern shows the curve about the boundary, the FD will be higher. Therefore, since benign tumours often have smooth boundaries and malignant tumours have uneven boundaries, this feature can be used as a distinction criterion.

2.2.4 Correlation dimension: CD is another chaos measurement tools to gain information about the nature of the underlying dynamics of a chaotic process in the form of time series. The dimension of a signal can describe the nature of the system. CD (D_2) is an index for measuring the complexity value of a dynamical system. CD is compatible with FD and related to the topological dimension of its attractor [36]. CD is related to the correlation integral, $C(r)$, described as following:

$$C(r) = \lim_{N \rightarrow \infty} \frac{1}{N(N-1)} \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \theta(r - |x_i - x_j|) \quad (10)$$

In (10), N is the attractor length (i.e. the number of data points), θ denotes the Heaviside function and r stands for the variable distances. Thus, $C(r)$ is the probability that the distance between pairwise points (x_i and x_j) is less than or equal to r in the attractor. The $C(r)$ is as follows:

$$C(r) \simeq ar^{D_2} \quad (11)$$

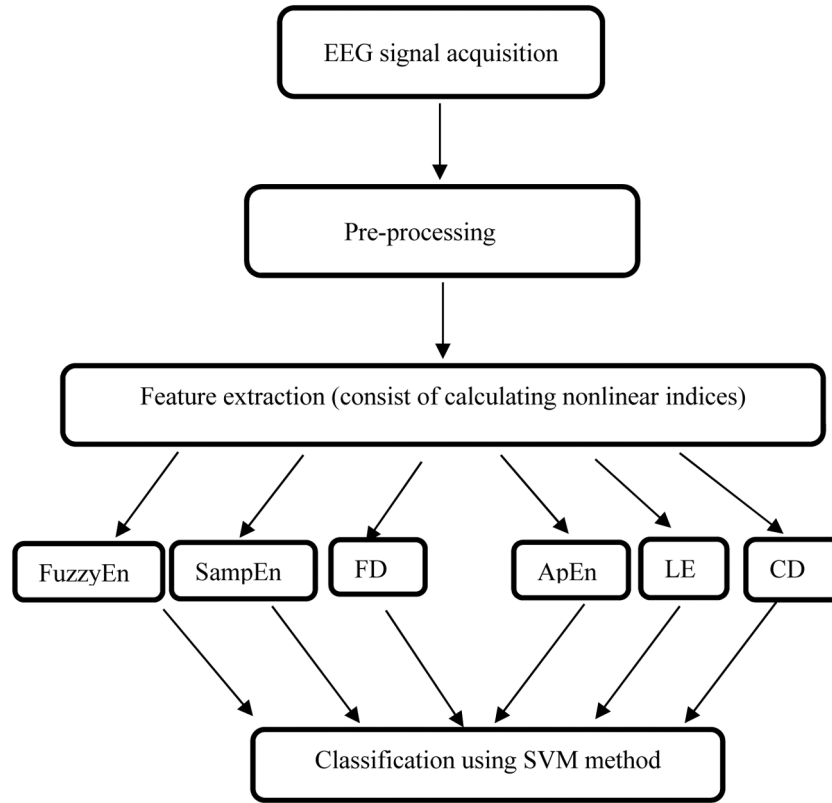


Fig. 4 Flow diagram of the proposed CAD method

where α is a fixed value, and D_2 is the computed slope for the $\log C(r)$ versus $\log r$ plot, defined as correlation exponent. Hence, the D_2 can be estimated as:

$$D_2 = \lim_{r \rightarrow 0} \frac{\log(C(r))}{\log(r)} \quad (12)$$

If increasing in the embedding dimension value leads to saturation for the CD, the system can be considered as a chaotic system [36].

2.2.5 Approximate, sample and FuzzyEn: Entropies are one of the most popular complexity measures used to analyze the biological signals [37, 38]. In this subsection, three widely used entropies named ApEn, SampEn, and FuzzyEn are discussed. ApEn can quantify the amount of irregularity in time series [37]. This technique reflects the signal overall characteristics and measures the time series complexity. Lower computational demand, applicable for small data samples and insensitivity to low-level noise makes this approach suitable for EEG signal analysis [39]. ApEn equation is as follows:

$$\text{ApEn}(m, r_i, N) = \Phi^m(r_i) - \Phi^{(m+1)}(r_i) \quad (13)$$

where

$$\Phi^m(r_i) = \frac{1}{N - m + 1} \sum_{i=1}^{N-m+1} \log C_i^m(r_i) \quad (14)$$

$$C_i^m(r_i) = \frac{\text{number of such } j \text{ that } d[x_m(i), x_m(j)] \leq r_i}{N - M + 1} \quad (15)$$

$$d[x_m(i), x_m(j)] = \max(|S(i + k - 1) - S(j + k - 1)|) \quad (16)$$

$$x_m(i) = \{S(i), S(i + 1), \dots, S(i + m - 1)\}; \quad 1 \leq i \leq N - m + 1 \quad (17)$$

In the above equations, m is data length and r_i is dedicated to filtering level. The sample numbers and the distance between two vectors $x_m(i)$ and $x_m(j)$ are indicated by N and d , respectively. The

small value (nearly zero) of ApEn shows that the signal has a regular nature [13].

SampEn is another method used for measuring the complexity degree of a time sequence. This method is developed on ApEn and has two important features. First, it is simpler than the ApEn algorithm, because it is largely independent to the length of record data and eliminates the self-matches. Second, the definition of SampEn is the generation rate of the new data. This approach can be used for analysing the short and noisy time series of clinical signals. The unpredictability of a signal can be indicated by a high value of SampEn while the similarity is displayed by its low value. Consistency, independence to the data length and sensitivity to small variations are the main features of SampEn [21].

FuzzyEn is an index used for estimating the rate of complexity in a time series [28]. The method used in this paper for estimating the FuzzyEn is similar to [40] in which, by assuming a time sequence, a vector is reconstructed using TDE method. Afterward, by calculation of the maximum distance between two neighbour states and assigning a fuzzy function, the similarity degree between two vectors is obtained and employed for computing the FuzzyEn.

2.3 Classification

Classification is one of the essential steps of medical data analysis in which unknown labels are predicted based on the observed feature [41]. SVM as one of the supervised learning models is employed in this study for EEG signals classification. Via finding a differentiating boundary named hyperplane, SVM classifies various cases. This algorithm, similar to the other classifiers, requires some training data. Through the training phase, a model is built that has the ability to determine new samples are belonged to which categories [12, 28, 34]. There are several choices for selecting the kernel function in which, the linear, polynomial, radial basis function and sigmoid are the most applicable.

When the classifying is performed, four indices consist of true positive (TP), true negative (TN), false negative (FN), and false positive (FP) are computed. These features indicate the sample numbers identified as positives or negatives; properly or incorrectly [28, 32, 34]. The obtained indices are used for calculation of some classification functions such as sensitivity, specificity, and accuracy. In (18), the evaluation terms are defined.



Fig. 5 Selected section of EEG signals without noise and artifacts in normal case



Fig. 6 Selected section of EEG signals without noise and artifacts in ADHD case

Table 1 Mean and standard deviation values of chaotic indices, calculated for normal and ADHD cases

Chaotic index	ADHD	Normal
LE	0.8844 ± 0.073	0.8504 ± 0.088
CD	0.7393 ± 0.033	0.7073 ± 0.031
FD	1.3677 ± 0.011	1.3550 ± 0.010
FuzzyEn	1.3261 ± 0.147	1.2642 ± 0.137
SampEn	2.4473 ± 0.095	2.3995 ± 0.092
ApEn	0.0125 ± 0.005	0.0095 ± 0.002

$$\begin{aligned}
 \text{Sensitivity} &= \frac{TP}{TP + FN} \\
 \text{Specificity} &= \frac{TN}{TN + FP} \\
 \text{Accuracy} &= \frac{TP + TN}{P + N}
 \end{aligned}
 \quad (18)$$

The flow diagram of a CAD system for ADHD diagnosis based on the proposed method is presented in Fig. 4. The four main steps of this system are the EEG signal acquisition, pre-processing, feature extraction, and classification.

3 Results

In this section, the proposed indices for extraction of EEG signals features have been used for the distinction between healthy

children and ADHD cases. The proposed method is evaluating on the obtained data set and the results are listed below.

3.1 Data acquisition

As previously mentioned, the EEG signals from 50 children with ADHD as well as 26 normal individuals were acquired in three equal modes, consist of EO, EC, and CT. Moreover, the placement of EEG electrode was compatible with the international 10/20 system and the brain activities were recorded in 19 regions. The provided data set is utilised for evaluating the ability of the proposed method.

3.2 EEG signal pre-processing

Due to less noise and artifact of EEG signals in closed eye condition rather than open eyes and cognitive tasks, we focused on this mode for achieving more accuracy in the calculations and the diagnosis. In addition, in the EC mode, the brain dynamics of ADHD patient can be analysed with respect to no external stimulus. Examples of EEG signals in normal and ADHD cases are illustrated in Figs. 5 and 6, respectively. The specified sections of these images, indicated by the pink color, are the selected seconds of signals has no noise and artifact. This selection is performed by a specialist and composed the visual inspection part of the pre-processing stage. Afterward, the selected seconds of EEG time series are passed with several filters to provide the artifact-free EEG signals for any individuals.

3.3 Feature extraction

The introduced non-linear indices of Section 2.2 are computed for the data set signals. Table 1 presents the average values of chaotic indices for normal and ADHD cases. The means of calculated chaotic indices consist of LE, FD, CD, FuzzyEn, SampEn, and ApEn of the ADHD group are significantly higher than the normal cases. The obtained significant differences could prove the chaotic nature of EEG signals in children with ADHD and illustrate the ability of the proposed approach in distinguishing two categories. Each of the calculated non-linear indices shows an aspect of the chaotic behaviour of the EEG signals. Sensitivity to the initial condition, pattern complexity, and irregularity of a signal can be measured by the introduced features.

3.4 Classification

To evaluate the potency of the obtained non-linear features in the distinction between control and ADHD groups, the recorded EEG signals need to be classified. For this purpose, SVM is applied. In total, 70% of the data set are utilised for training the classifier while the remaining 30% are employed for testing. Table 2 presents the results of computing the evaluation terms for several combinations of non-linear features in Fz channel. The obtained results are the best accuracies achieved through the polynomial kernel for SVM.

According to Table 2, using the combination of CD and FD, or combination of CD and SampEn as the selected features and SVM as the classifier yield to a 98% sensitivity, 92.31% specificity, and 96.05% accuracy for Fp2 and Fz channel, respectively. These results are the best calculated for all channels. In addition, the best results of classification performance in terms of selected non-linear features for each channel are calculated and shown in Table 3.

It can be seen that the selection of CD and SampEn in Fz channel and CD and FD in Fp2 channel lead to the highest percentage of sensitivity, specificity, and accuracy, in diagnosis and classification of ADHD and control groups, compared to other calculated channels.

4 Discussion

For evaluating the capability of the proposed method in the distinction between normal and ADHD cases, the obtained results in the current study are compared with other similar works, in which, the proposed method has been implemented in a data set with at least 60 cases. Table 4 shows the performance of the

Table 2 Best obtained results based on the combination of various chaotic indices

Chaotic indices	Sensitivity, %	Specificity, %	Accuracy, %	Frontal lobe
LE, SampEn	94	73.08	86.84	Fz
LE, CD	96	76.92	89.47	Fz
LE, FuzzyEn	92	73.08	85.53	F3
LE, FD	96	76.92	89.47	F3
LE, ApEn	90	65.38	81.58	Fz
CD, FD	98	92.31	96.05	FP2
CD, FuzzyEn	96	84.62	92.11	F3
CD, SampEn	98	92.31	96.05	Fz
CD, ApEn	94	69.23	85.53	FP2 and Fz
FD, FuzzyEn	96	69.23	86.84	F4
FD, SampEn	92	69.23	84.21	Fz
FD, ApEn	88	76.92	84.21	FP2 and Fz
ApEn, SampEn	92	76.92	86.84	F3
ApEn, FuzzyEn	94	57.69	81.58	Fz
FuzzyEn, SampEn	96	73.08	88.16	Fz

Table 3 Best results for each of the frontal regions of the brain

Frontal lobe	Selected chaotic indices	Sensitivity, %	Specificity, %	Accuracy, %
FP1	CD, FD	92	92.31	92.11
FP2	CD, FD	98	92.31	96.05
F3	CD, FD, CD, SampEn	96	84.62	92.11
F4	CD, FD	96	84.62	92.11
Fz	CD, SampEn	98	92.31	96.05

Table 4 Comparison between accuracies of this approach and the published results

Authors	Year	Number of subjects	Age	Measures	Accuracy, %
Williams <i>et al.</i> [42]	2010	169 ADHD and 167 control	7–19	Fz/FCz, Θ/β ratio	($p < 0.05$)
Loo <i>et al.</i> [43]	2013	390 ADHD and 100 control		Cz, Θ/β ratio	38
Liechti <i>et al.</i> [44]	2013	54 ADHD and 51 control	8–16	Cz, Θ/β ratio	53
Buyck and Wiersema [29]	2014	62 ADHD and 55 control	5–15 and 20–50	Cz, Θ/β ratio	49–55
Helgadottir <i>et al.</i> [45]	2015	90 ADHD and 90 control	6.5–8.2	multi-channels, optimal features	79
		90 ADHD and 90 control	10.4–12		
		90 ADHD and 90 control	12.2–14		
		45 ADHD and 45 control	6.5–14		
Tenev <i>et al.</i> [13]	2013	67 ADHD and 50 control	18–50	Multi-channel, absolute power ($\delta\Theta\alpha\beta\gamma$)	82.3
Ogrim <i>et al.</i> [46]	2012	62 ADHD and 36 control	7–16	Cz, absolute Θ and β power Θ/β ratio	85
Markovska-Simoska and Pop-Jordanova [16]	2016	30 ADHD and 30 control	6–14	Cz, Θ/β ratio	87.9
Snyder <i>et al.</i> [47]	2008	97 ADHD and 62 control	6–18	Cz, Θ/β ratio	89
Mohammadi <i>et al.</i> [14]	2016	30 ADHD and 30 control	7–14	multi-channel, non-linear optimal features	93.65
This paper		50 ADHD and 26 control	4–15	multi-channel, non-linear indices	96.05

combination of selected non-linear indices is completely satisfactory in comparison to the other methods which are utilised EEG data set. The yielded improvement illustrates the potency of non-linear features in characterising the chaotic behaviour of ADHD based on the relative EEG signals. Although the EEG time series are recorded in three modes consist of EC, EO, and CT, the EC is selected for signal analysing. Less effect of noise and artifact on EEG signals and no external stimulus on brain dynamics in EC condition, caused the analysing of the EC mode signals leads to higher accuracy.

The obtained results demonstrate that the selected non-linear indices are suitable tools for characterising the EEG signals. Hence, the chaotic quantifying methods can characterise the significant properties of such a complex signal. It can comprehend that more complexity existed in the brain system of ADHD cases leads to larger calculated values for non-linear indices namely LE, SampEn, FuzzyEn, ApEn, CD, and FD. As previously mentioned, each of non-linear indices shows an aspect of the chaotic behaviour of the EEG signals. Therefore, their combination can represent the chaotic nature of the EEG signals in ADHD cases.

5 Conclusion

ADHD is known as a common disorder in children and if diagnosed in early stage, its future problems will be decreased. In the current study, a CAD system is presented by using the chaotic analysis of multi-channel EEG signals for ADHD diagnosing. Besides, the SVM is employed as a classifier to increase diagnostic accuracy.

In addition, experimental results on an obtained data set of EEG signal related to ADHD and healthy individuals illustrate the ability of the proposed method in distinguishing the normal and abnormal EEG signals. The remarkable precision of the selected non-linear features compared to other works indicates their acceptable capability in the classification of the chaotic and deterministic signals, which is a crucial factor in the diagnosis of ADHD children.

It should be noted that the CADs can be assumed as a medical procedure that helps physicians to interpret the medical signals and images. Through the CAD, neurologist and radiologists utilise the computer outputs as 'second opinions' to adopt the final decision. Hence, the calculated results for ADHD diagnosing by the

proposed method enhance doctor's competence during the medical examination.

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